PATIENT CARE THEORY 2

UNIT 5, PART 4: Labour & Delivery

Marilyn Niffin, BSc, ACP
Professor Georgian College
Patient Care Theory 2, Winter 2020, 2021 (online)

Learning Objectives

- Identify the stages of labour and the events that occur in each phase
- Discuss patient management for each stage of labour
- Differentiate between true labour and false labour
- Explain the procedure for normal delivery of a neonate
- Discuss management of both mother and neonate following delivery
- Discuss the midwife – paramedic relationship and role expectations

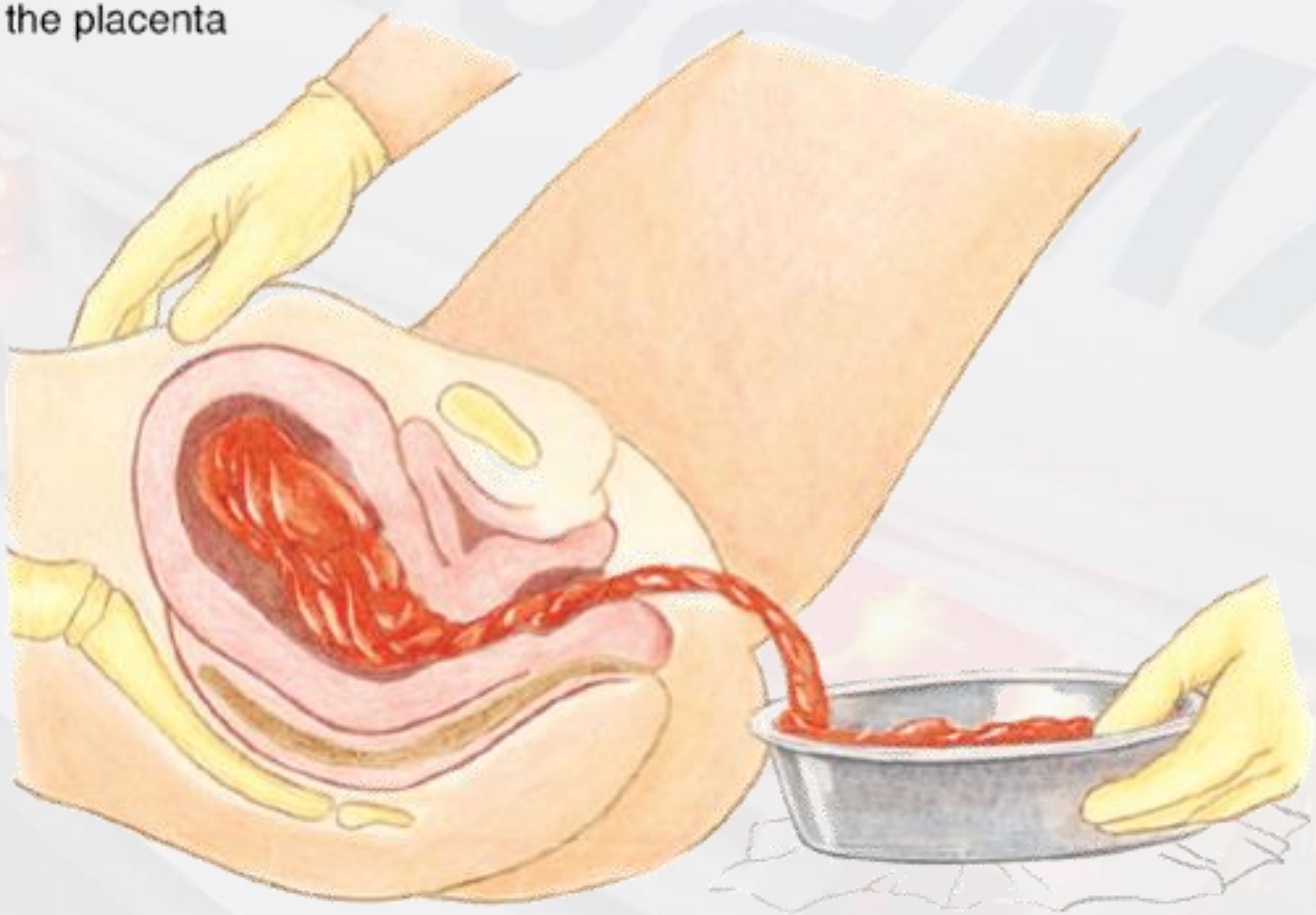
Labour

First stage of labour - Transport

First stage: beginning of contractions to full cervical dilation

Second stage of labour – Stay & Deliver

Third stage of labour - Transport



Labour Synopsis

- Stage 1 (Dilation)
  - Onset of true contractions to full dilation and effacement of cervix (thinning of the cervix - expressed in percentages. 100% effacement = begin pushing)
- Stage 2 (Expulsion)
  - Full dilation to delivery of fetus
- Stage 3 (Placental)
  - Immediately following the delivery of the fetus to delivery of the placenta

First Stage of Labour

- Described as “early labour” and “active labour”
- Usually the longest of the 3 stages
- Gradual increase of uterine contractions (true contractions begin) – frequency, length and strength over time
- Contractions
  - Early – mild and irregular – unpredictable and can occur on and off over days
  - Early effacement and dilation can result in discharge (pink, red or brown)
  - SROM may or may not occur

First Stage of Labour

- Fetus descends in to pelvis
- Cervix softens, shortens, effaces (thins), and dilates

- Paramedic Observations:
  - Contractions far apart and irregular
  - Length – usually short (20-30 secs)
  - Mild strength
- Patient Observations:
  - Walking and talking through contractions

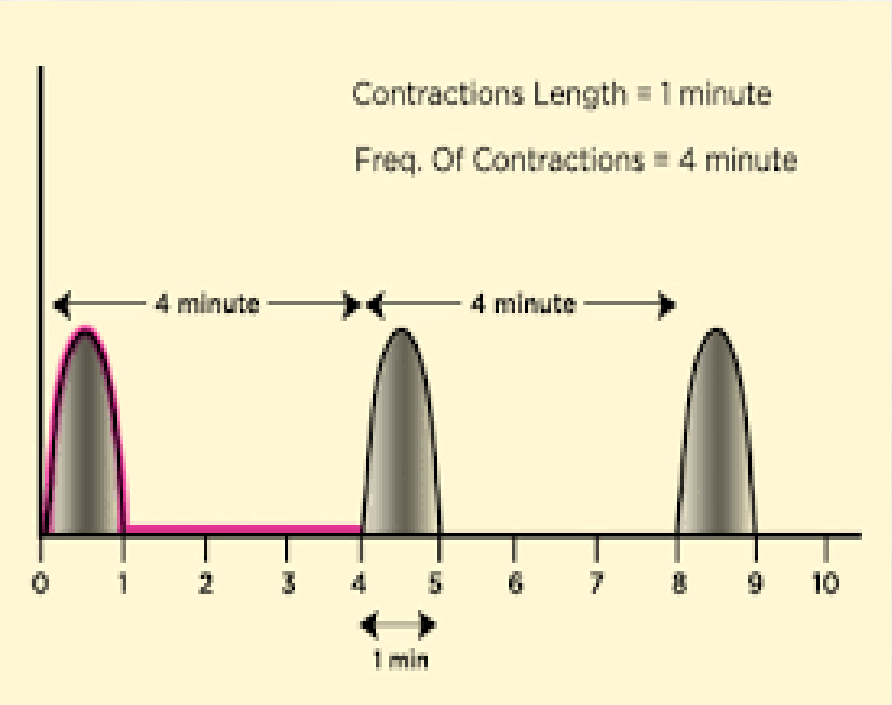
First Stage of Labour

- Active labour
  - Cervical changes continue
    - “bloody show”
    - SROM may or may not occur
  - Contractions – more regular, (3-4 minutes apart, lasting 30-40 secs)
  - Mild –moderate strength
- Patient Behaviour
  - Difficulty walking and talking through contractions
  - May see vomiting, shaking, increase in bloody show, overwhelming emotions, SROM (if not already)

First Stage of Labour

- Primips
  - Active labour on avg 8 hrs – unlikely more than 18 hrs
- Multips
  - Active labour on avg 5 hrs – unlikely more than 12 hrs
- Management
  - BLS standards
  - Determine Active or early labour (time contractions)
  - Load and go

First Stage of Labour



- How are contractions "timed"
- Frequency – length of time from the beginning of one contraction to the beginning of the next
- Length – duration of the contraction
- Strength – how tight is the muscle at the peak of the contraction (fundus)

Second Stage of Labour

- Begins when cervix fully dilated
- Patient feels “urge to push” or have a BM
- Complete when baby is born

- Paramedic observations:
  - Uterus will “rise up” with contraction
  - Patient will bear down with contractions
  - Blood, mucous, fluid may be draining
  - Increased pain with stretching of perineal tissues
  - Bulging or distension may be seen
  - Presenting part may be visible

Second Stage of Labour

- Signs of imminent birth
  - Strong contractions (60-90 secs long)
    - Primips 2-3 min apart; multips 5 min apart
  - Cramping/pain that impedes ambulation
  - Pain in buttocks, pressure or burning in perineum
  - Urge to push, “baby is coming”
  - Bulging membranes/perineum

Prepare for the OOH birth

- Attempt to provide a quiet, safe, warm, private environment
- Position the patient supine, with her head raised, legs flexed and abducted at knees

Management

- Delivery is imminent when…
  - Crowning
  - primips:
    - presenting part visible during and between contractions;
    - urge to push or bear down; restless;
    - contractions 2-3 minutes, or
  - Multips:
    - contractions ≤ 5 minutes apart + any other signs of second stage labour
      - (urge to push or bear down;
      - heavy red show;
      - presenting part or bulging membranes visible at the vaginal orifice during or between contractions
  - Water is broken, strong contractions and patient actively pushing

FIGURE 40-8 Crowning.

Prepare for OOH Birth

- Attempt to maintain privacy
- Ensure patient is on a firm flat surface
- Position supine with head elevated, legs abducted at hip and knees (if possible – patient may request alternate position)
- Gather equipment:
  - OBS kit, PPE
  - Additional blankets/sheets/ towels for newborn
  - Place sterile sheet under buttocks of patient
  - Attempt to maintain warmth

Crowning

FIGURE 40-8 Crowning.

- support the head to prevent explosive birth and tearing of the perineum
- You may need to stretch the perineum
- Be prepared for tearing and bleeding
- **Document colour of fluid if possible (take a mental note or ask partner to document)**

Delivery of the head

- Document the time the head is delivered
- Check for nuchal chord
  - If present and loose, slip over the baby's head
  - If too tight and unable to slip over head or reduce tension, clamp and cut the cord (patching at this point is recommended)
- Document the presence of a nuchal cord and actions taken

FIGURE 40-9 Delivery of the head.

External rotation of the head

- Routine suctioning of baby is NOT RECOMMENDED at this point.
- If mucous or meconium is present, wipe away
- Encourage patient to push with next contraction

FIGURE 40-10 External rotation of the head.

Deliver the **anterior** shoulder

- Apply gentle downward pressure to deliver the anterior shoulder
- Followed by upward flexion to deliver the posterior shoulder
- Prepare for the baby to come quickly at this point

Complete the delivery of the infant

- Document the time of delivery

Place the infant on the mother's abdomen

FIGURE 40-15 Place the infant on the mother's abdomen.

- This allows for warmth
- Dry, Stimulate and assess the newborn (for tone, breathing and crying)
- Assess APGAR at the one and five minute marks

Clamp and cut the cord

- If no resuscitation is required, wait 2-3 minutes until pulsations have stopped.
- Place clamps in 2 locations (approximately 15 cm from the newborn's abdomen then another 5 cm above that; cut between the 2 clamps

Dry & warm the infant
re-assess airway

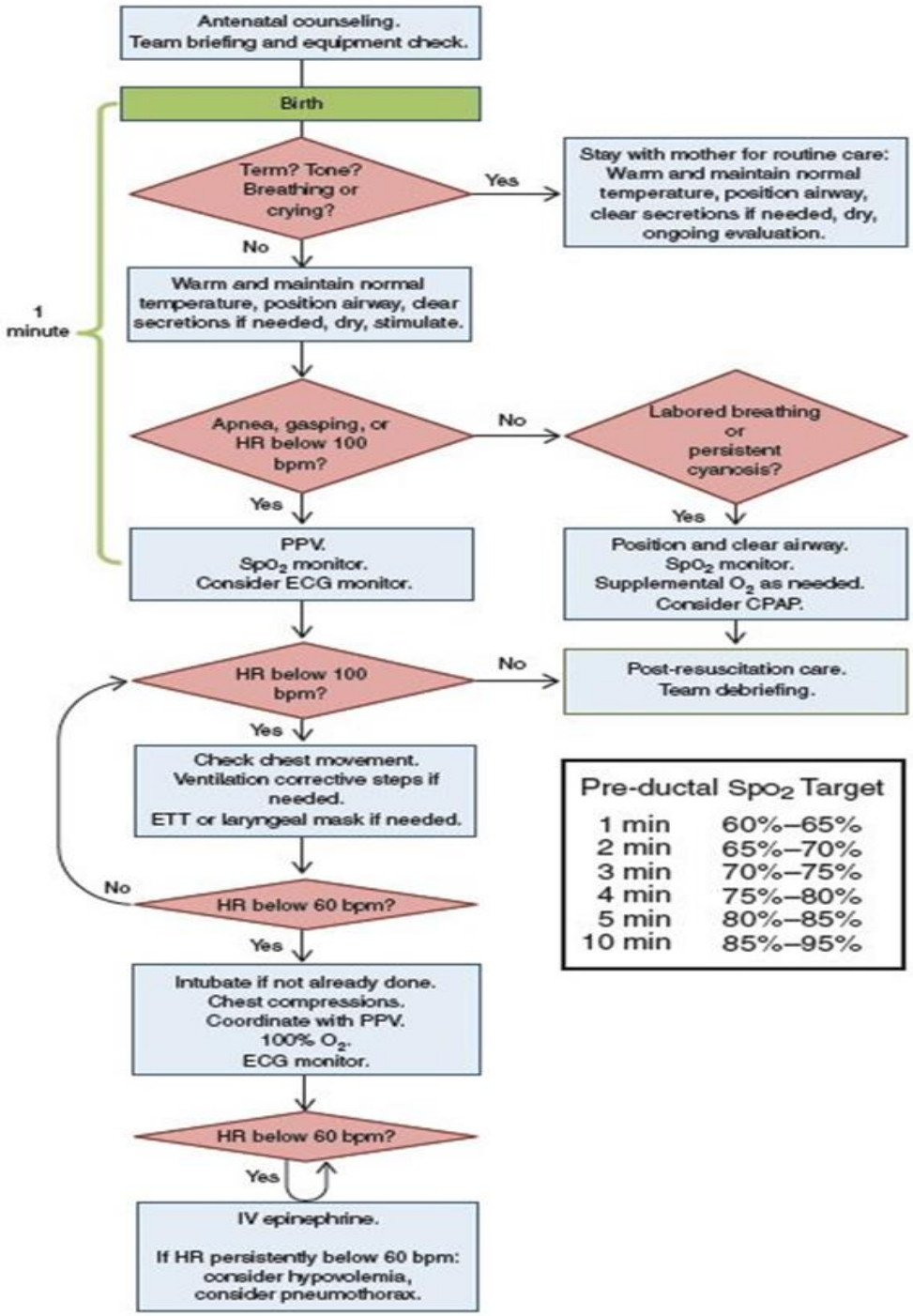
Be aware that the baby may not cry immediately. Stimulation may be required

What impact can the position of the baby have on it's circulating blood volume [prior to clamp & cut of the cord]?

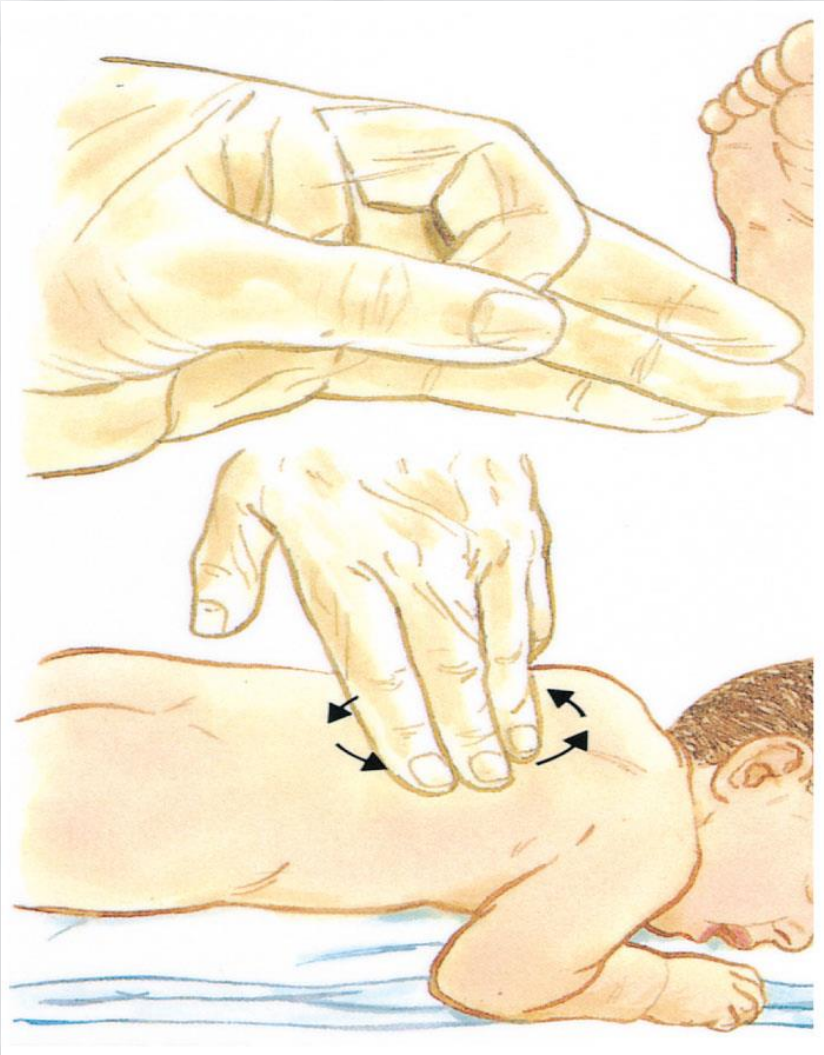
Answer:

Neonatal Care

- Support the infant's head and torso, using both hands
- Dry newborn (extremely susceptible to hypothermia)
- Maintain warmth
- ABC re-assessment: Only if needed, suction mouth, then nose
  - Newborns are "obligate nose breathers"
    - What does this mean?


Antenatal counseling.
Team briefing and equipment check.
Birth
Term? Tone? Breathing or crying?
Yes
Stay with mother for routine care: Warm and maintain normal temperature, position airway, clear secretions if needed, dry, ongoing evaluation.
No
1 minute
Warm and maintain normal temperature, position airway, clear secretions if needed, dry, stimulate.
Apnea, gasping, or HR below 100 bpm?
No
Labored breathing or persistent cyanosis?
Yes
PPV.
SpO_2 monitor.
Consider ECG monitor.
Yes
Position and clear airway.
SpO_2 monitor.
Supplemental O_2 as needed.
Consider CPAP.
HR below 100 bpm?
No
Post-resuscitation care.
Team debriefing.
Yes
Check chest movement.
Ventilation corrective steps if needed.
ETT or laryngeal mask if needed.
No
HR below 60 bpm?
Yes
Intubate if not already done.
Chest compressions.
Coordinate with PPV.
100% O_2.
ECG monitor.
HR below 60 bpm?
Yes
IV epinephrine.
If HR persistently below 60 bpm: consider hypovolemia, consider pneumothorax.
Pre-ductal SpO_2 Target
1 min 60%–65%
2 min 65%–70%
3 min 70%–75%
4 min 75%–80%
5 min 80%–85%
10 min 85%–95%

Stimulate the infant as required



Neonatal Care

- Assess the neonate using Apgar score (Performed at 1 minute and 5 minutes after birth

Table 40-2 THE APGAR SCORE

Element	0	1	2	Score
Appearance (skin color)	Body and extremities blue, pale	Body pink, extremities blue	Completely pink	
Pulse rate	Absent	Below 100/min	100/min or above	
Grimace (Irritability)	No response	Grimace	Cough, sneeze, cry	
Activity (Muscle tone)	Limp	Some flexion of extremities	Active motion	
Respiratory effort	Absent	Slow and irregular	Strong cry	
			TOTAL SCORE =	

Neonatal Care

Is the newborn "vigorous" ?

- ✓ *good muscle tone*
- ✓ *strong respiratory effort*
- ✓ *heart rate $\geq$ 100 (how do we check this?)*

Post-Partum – Assuming no resuscitation req'd

- Skin to skin is best
- Allow/Encourage Mom to breast feed
  - Stimulates the release of oxytocin
    - hormone secreted by the pituitary gland
    - stimulates contractions of the uterus and the milk-eject reflex
    - Helps controls post-partum bleeding
- Extremely important to maintain heat. Hypothermia significantly increases the risk of post-partum complications
- Administer Oxytocin

Medical Directive – ALS PCS 5.0

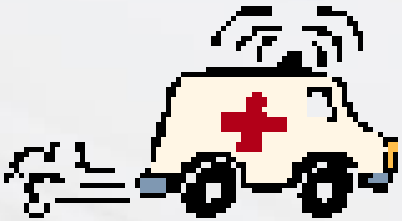
Conditions

Oxytocin	
Age	Childbearing years
LOA	N/A
HR	N/A
RR	N/A
SBP	≤ 160 mmHg
Other	Postpartum delivery **AND/OR** Placental delivery

Contraindications

Oxytocin
Allergy or sensitivity to oxytocin
Undelivered fetus
Suspected or known pre-eclampsia with current pregnancy
Eclampsia (seizures) with current pregnancy
>4 hours post placenta delivery

Go



Follow Service Policy for transportation of Newborn baby and mother

THIRD STAGE OF LABOUR

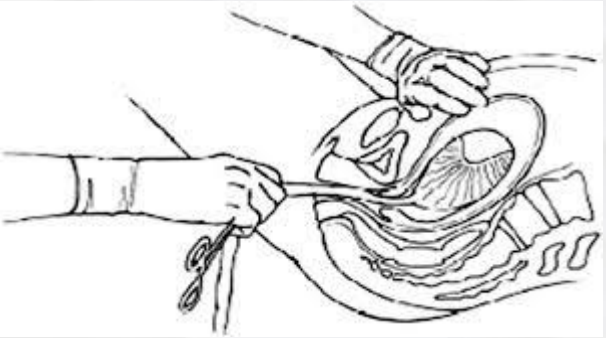
Delivery of the placenta – usually en route

THIRD STAGE OF LABOUR

- The separation of the placenta and membranes from the uterus
- Typically occurs 5-30 min after birth
- Paramedic Observations:
  - Lengthening of the cord
  - Sudden “gush” or trickle of blood from the vagina
  - Uterine contractions

THIRD STAGE OF LABOUR

- Management
  - Ask the patient to push
  - Apply gentle controlled cord traction (CCT)
    - One hand holds the cord and applies CCT while the other hand is placed just above the pubic bone and guards the uterus



THIRD STAGE OF LABOUR

- Document:
  - Time of placental delivery
  - Estimated amount of blood loss/clots
  - Inspect placenta for wholeness
    - Place it in a bag (provided in OB kit), label and bring with the patient to the receiving facility
  - Monitor mother for signs of PPH
    - Perform uterine massage if PP bleeding still present (see ALS PCS)

Midwife on Scene

- Midwives are a primary care regulated health profession(College of Midwives of Ontario)
- If there is a Midwife on scene, they are considered the higher medical authority in this case (for the delivery)
- Work together for the optimal care of the mother and the baby
- Transport may or may not be required –
  - Consideration for location may be based on where the midwife has privileges and/or patient condition

QUESTIONS